



ICT 1201

Fundamentals of Computer Systems

The Computer System

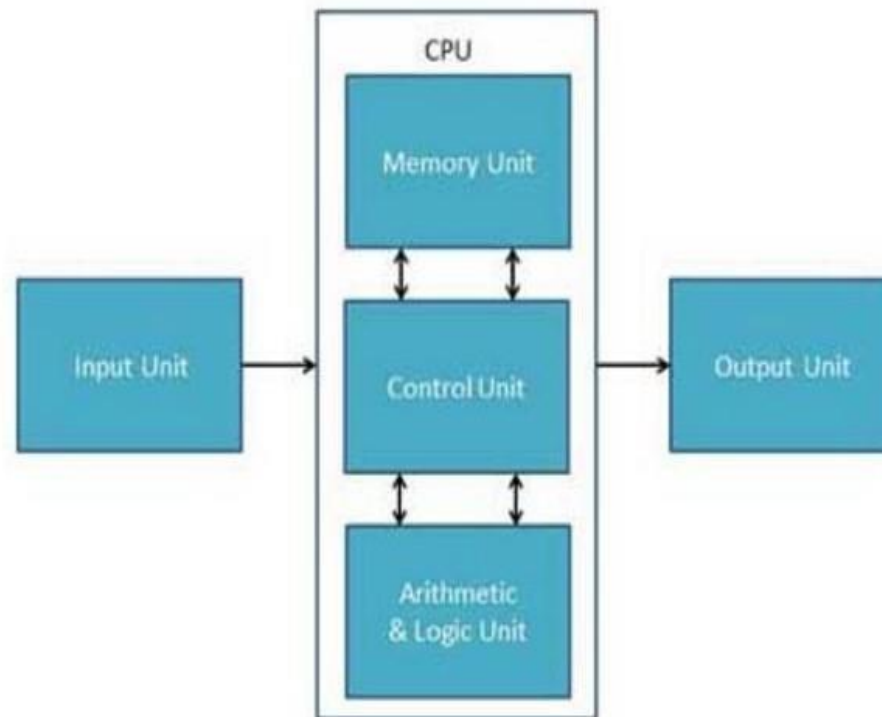
A Typical Computer



FIGURE 1-7
Typical computer hardware.

Components of a Computer

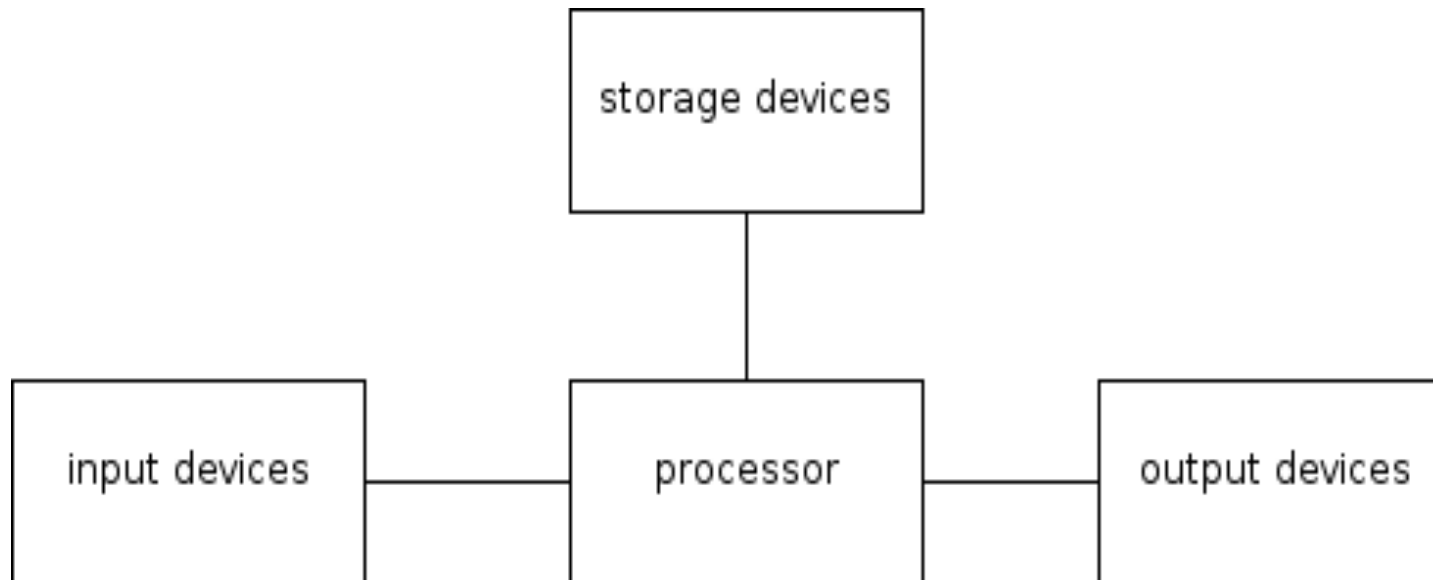
All types of computers follow a same basic logical structure and perform the following five basic operations for converting raw data into information useful to their users.



Functions of a Computer

Operation	Description
TakeInput	• The process of entering data and instructions into the computer system.
Store Data	• Saving data and instructions so that they are available for processing as and when required.
Processing Data	• Performing arithmetic and logical operations on data in order to convert them into useful information.
Output Information	• The process of producing useful information or results for the user, such as a printed report or visual display.
Control the workflow	• Directs the manner and sequence in which all of the above operations are performed.

Basic Components of a Computer



Input unit

- ❑ This unit contains devices with the help of which we enter data into computer.
- ❑ This unit makes link between user and computer.
- ❑ The input devices translate the information into the form understandable by computer.

Storage Devices

- ❑ Components which allow data to be stored within a computer system.
- ❑ This includes devices such as hard disk drives and compact drives.



Output Unit

- ❑ Consists of devices with the help of which we get information from computer.
- ❑ This unit is a link between computer and users.
- ❑ Output devices translate the computer's output into the form which can be understood by users.

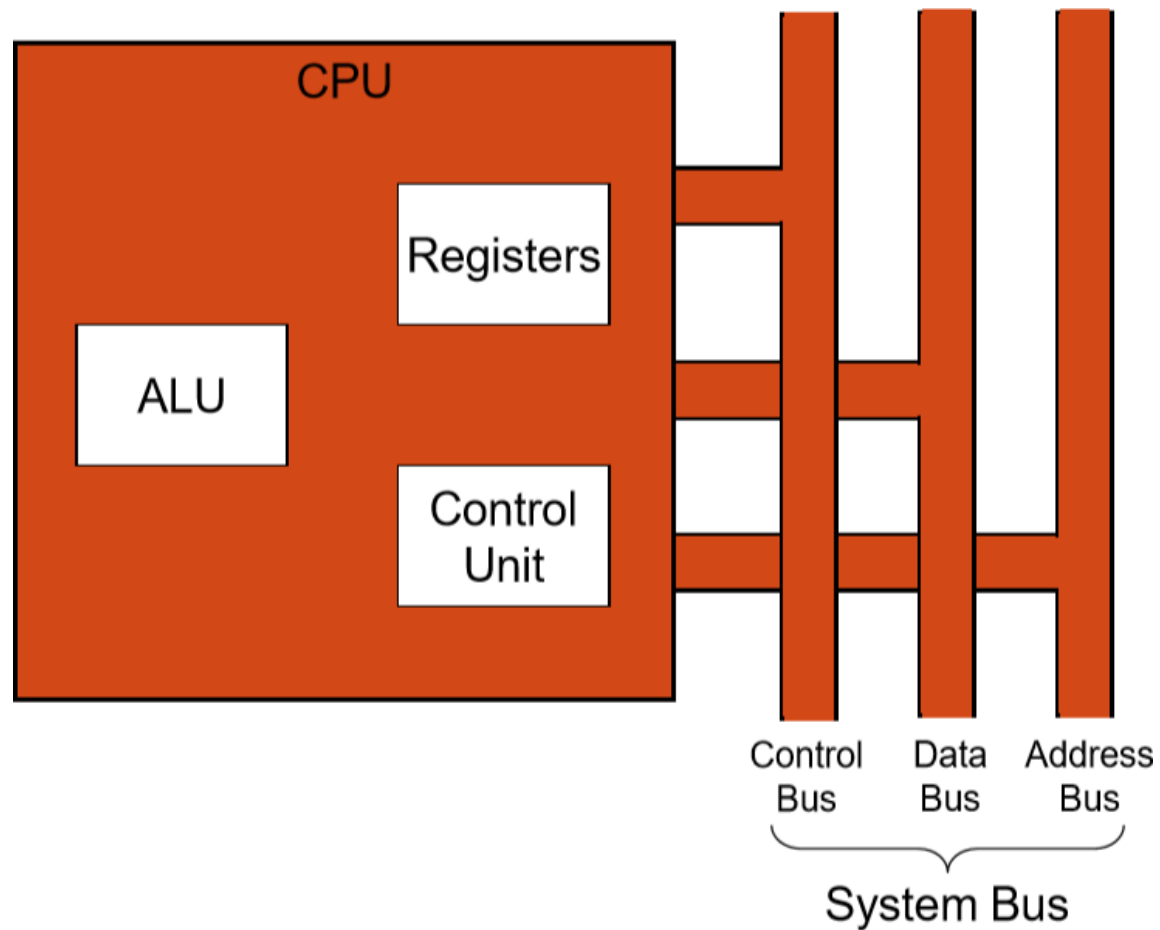
CPU Functions

- ❑ Fetch instructions.
 - CPU must read instructions from memory.
- ❑ Interpret instructions.
 - Instructions must be decoded to determine what action is required.
- ❑ Fetch data.
 - Execution of an instruction may require reading data from memory of an IO module.
- ❑ Process data.
 - Execution of an instruction may require performing some arithmetic or logical operation on data.
- ❑ Write data.
 - Results of an instruction execution may require writing data to memory or an IO module.

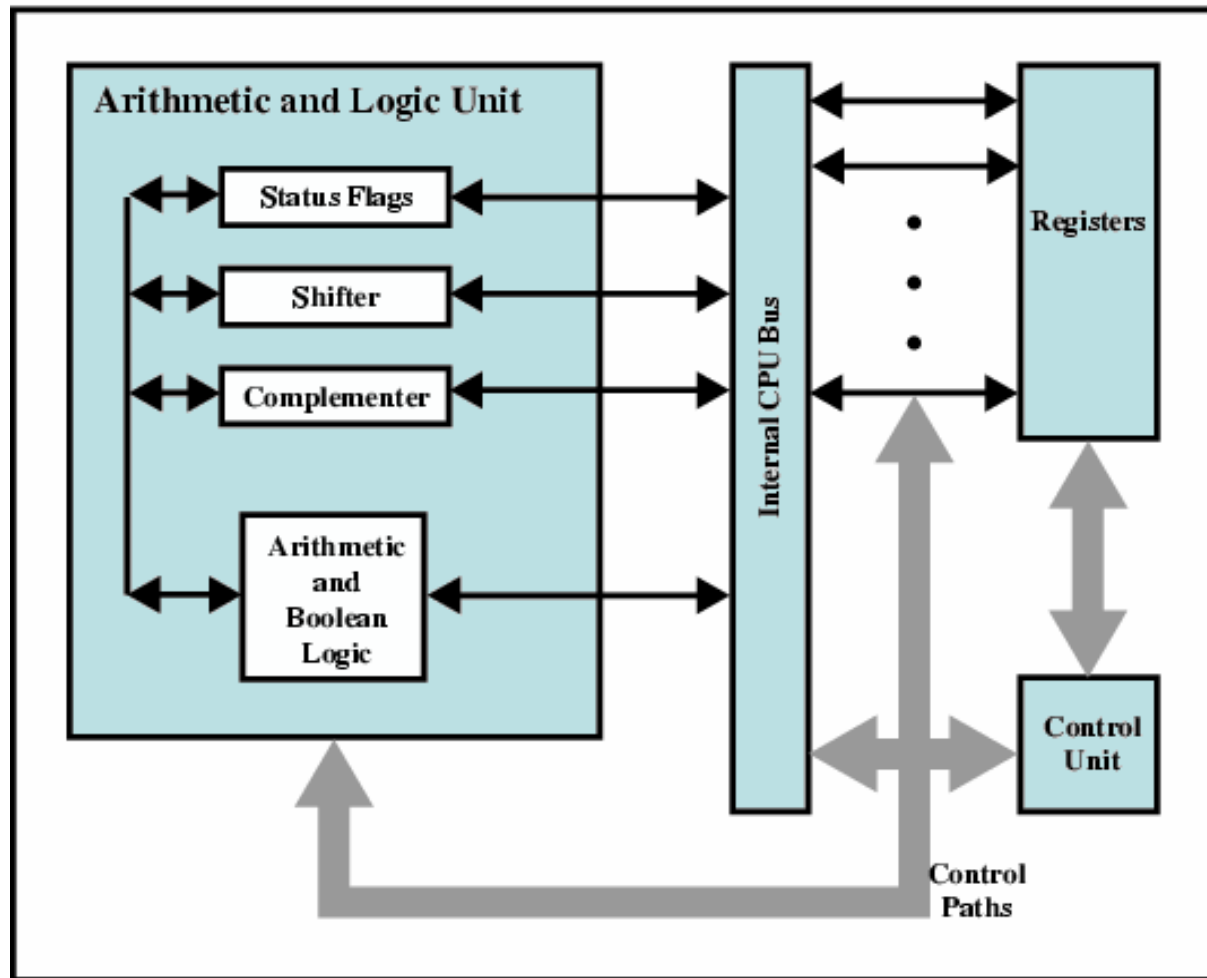
Main Components of the CPU

- ❑ Control Unit(CU)
 - Controls the operation of the CPU and hence the computer.
- ❑ Arithmetic and Logic Unit (ALU)
 - Performs computer's data processing functions.
- ❑ Registers
 - Provides storage internal to the CPU.
 - High speed memory locations.
- ❑ CPU Interconnections(BUS)
 - Some mechanism that provides for communication among the control unit, ALU and registers.

Simplified view of a CPU



Internal Structure of the CPU



ALU (Arithmetic Logic Unit)

- ❑ This unit consists of two subsections namely
 - Arithmetic Section
 - Perform arithmetic operations like addition, subtraction, multiplication and division. All complex operations are done by making repetitive use of above operations.
 - Logic section
 - Perform logical operations such as comparing, selecting, matching and merging of data.

Control Unit

- ❑ This unit controls the operation of all parts of computer but does not carry out any actual data processing operations.
- ❑ Functions of this unit are:
 - It is responsible for controlling the transfer of data and instructions among other units of a computer.
 - It manage and coordinated all the units of the computer.
 - It obtains the instructions from the memory, interprets them, and directs the operation of the computer.
 - It communicates with Input/output devices for transfer of data or results from storage.
 - It does not process or store data.

CPU Registers

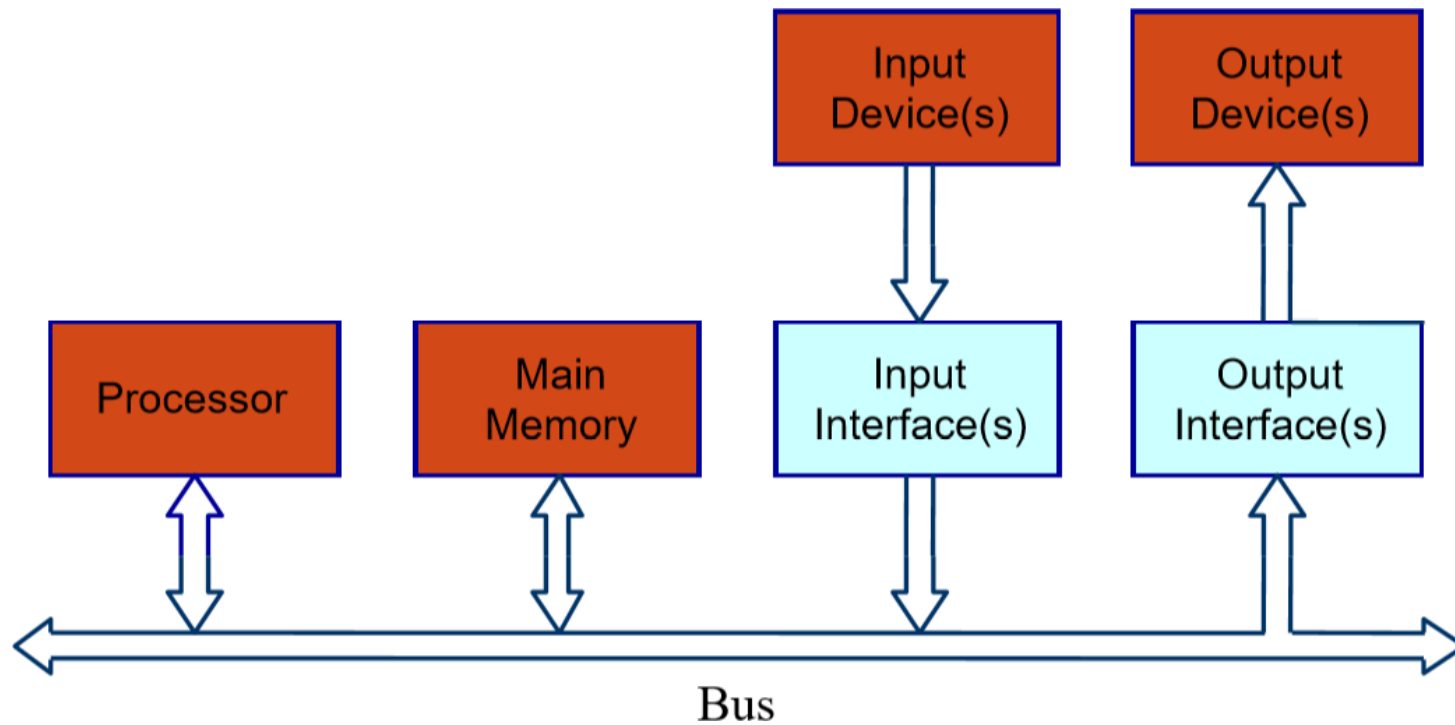
- ❑ CPU needs to store instructions and data temporarily while an instruction is being executed.
 - To remember the location of the last instruction.
 - To remember the location of the next instruction.
 - To hold the operands and results.
- ❑ CPU needs a small internal memory.
- ❑ Called Registers.
 - Set of storage locations.
 - Faster, Smaller and more expensive.
 - Number and function vary between processor designs.

System Interconnections

❑ Bus

- Communication pathway connecting two or more devices.
- Shared transmission (transportation) media.
 - Multiple devices are connected to a bus.
 - Only one device can transmit data at a time.
- A group of parallel wires.
 - Each wire in a bus can transfer one bit of information.
 - The number of parallel wires in a bus is equal to the word length of a computer (8 bits, 16 bits, 32 bits, 64 bits)
 - Bus can transfer word of information at a time.

System Interconnections



System Bus

- ❑ Computer system contain a number of different buses.
- ❑ A bus that connects major components (CPU, Memory, I/O) is called a System Bus.
- ❑ On any bus, the lines can be classified into three functional groups.
 - Data Lines (DataBus)
 - Address Lines (Address Bus)
 - Control Lines (ControlBus)

Data Bus

- ❑ Provide a path for moving data between system modules.
- ❑ No difference between “data” and “instruction” at this level.
- ❑ The data bus typically consists of 6, 16, or 32 separate lines.
- ❑ Each line carry only 1 bit at a time, the number of lines determine how many bits can be transferred at a time – overall system performance.
- ❑ Width is a key determinant of performance. (Width = Number of lines.)

Address Bus

- ❑ Identify the source or destination of data on the data bus.
- ❑ Bus width determines maximum memory capacity of the system.
 - E.g. 8080 has 16 bit address bus giving 64k address space (memory size)
- ❑ Also used to address I/O ports.

Control Bus

- ❑ Used to control the use of data and address bus.
- ❑ As a shared transmission medium.
 - E.g. BusRequest, BusGrant etc.
- ❑ Transfers control and timing information.
 - Memory read/write signal.
 - Interrupt request.
 - Clock signals.
 - I/O read write.

Instruction Set

- ❑ Instruction Set is the complete collection of different instructions that are understood by a CPU.
- ❑ Different Instruction Sets for different CPU designs.
- ❑ A large part of the CPU design involves implementing the machine instruction set, referred to as Processor's Instruction Set.

Instruction Set

- ❑ Instructions are
 - Machine Codes.
 - Binary Numbers (Stored in the Memory).
 - Usually represented by assembly codes.

Address	Contents				Label	Operation	Operand
101	0010	0010	0000	0001	FORMUL	LDA	I
102	0001	0010	0000	0010		ADD	J
103	0001	0010	0000	0011		ADD	K
104	0011	0010	0000	0100		STA	N
201	0000	0000	0000	0010	I	DAT	2
202	0000	0000	0000	0011	J	DAT	3
203	0000	0000	0000	0100	K	DAT	4
204	0000	0000	0000	0000	N	DAT	0

Binary Program

Assembly Program

Instruction Set

- ❑ In machine code each instruction has a unique bit pattern.
 - E.g. 1010100010100011
- ❑ For human use, a symbolic representation is used.
 - E.g. ADD, SUB, LOAD
- ❑ Operands can also be represented in the same way.
 - E.g. ADD A, B

Instruction Set

- ❑ Each instruction must contain four basic pieces of information.
 - Operation Code(Op Code)
 - Operation to be performed (Do this)
 - E.g. ADD, SUB, LOAD, STOR etc.
 - Source Operand Reference
 - Operands (inputs) for the operation (To this)
 - Result Operand Reference
 - Result of the operation (Put the answer here)
 - Next Instruction Reference
 - Next instruction to be fetched (When you have done that, do this...)

Hardware

- ❑ A generic term used to describe any component of a computer system with a physical presence and which can, therefore, been seen and touched.
- ❑ Hardware components are often categorized as;
 - Input
 - Output
 - Storage
- ❑ Devices which are not an integral part of the CPU are referred to as peripherals.

Input Devices

- ❑ Text entry devices.
- ❑ Positioning, pointing and drawing devices.
- ❑ Scanning devices.
- ❑ 3D interaction devices

Input Devices

❑ Text entry devices.

➤ Keyboard

- Most common text input device.
- The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provide for performing additional functions.
- Keyboards are of two sizes 8 keys or 101/102 keys, but now keyboards with 104/108 keys are also available.
- Key press closes connection, causing a character code to be sent.



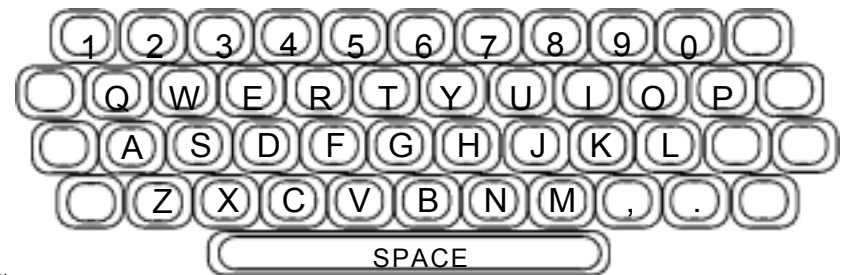
Input Devices

❑ Text entry devices.

➤ Keyboard

- Usually connected by cable, but can be wireless.
- There are several types available.

- QWERTY
- Alphabetic
- Dvorak
- Chord keyboards
- Special keyboards



Input Devices

❑ Text entry devices.

➤ Keyboard

■ The keys on the keyboard are as follows

Keys	Description
Typing Keys	These keys include the letter keys (A-Z) and digit keys (0-9) which generally give same layout as that of typewriters.
Numeric Keypad	It is used to enter numeric data or cursor movement. Generally, it consists of a set of 17 keys that are laid out in the same configuration used by most adding machines and calculators.
Function Keys	The twelve function keys are present on the keyboard which are arranged in a row at the top of the keyboard. Each function key has unique meaning and is used for some specific purpose.
Control keys	These keys provide cursor and screen control. It includes four directional arrow keys. Control keys also include Home, End, Insert, Delete, Page Up, Page Down, Control(Ctrl), Alternate(Alt), Escape(Esc).
Special Purpose Keys	Keyboard also contains some special purpose keys such as Enter, Shift, Caps Lock, Num Lock, Space bar, Tab, and Print Screen.

Input Devices

- ❑ Positioning, pointing and drawing.

- The Mouse

- Famous cursor-control handheld pointing device.
 - Generally it has two buttons called left and right button and a wheel is present between the buttons.
 - Movement of mouse moves screen cursor.

Input Devices

- ❑ Positioning, pointing and drawing.

- The Mouse

- Two types of mouse

- Mechanical

- » This uses a rolling ball to detect movement. Less expensive and it may require regular cleaning to keep it working properly.



Input Devices

- Positioning, pointing and drawing.

- The Mouse

- Two types of mouse

- Optical

- » Light emitting diode on under side of mouse.

- » Less susceptible to dust and dirt



Input Devices

- ❑ Positioning, pointing and drawing.

- Touchpad

- A touchpad also called a track pad is a touch sensitive pad that lets you control the pointer by making a drawing motion with your finger. This is common on laptop computers.



Input Devices

- ❑ Positioning, pointing and drawing.

- Light pen

- Pointing device which is similar to a pen.
 - Used to select a displayed menu item or draw pictures on the monitor screen.
 - It consists of a photocell and an optical system placed in a small tube.
 - When the tip of light pen is moved over the monitor screen and pen button is pressed, its photocell sensing element detects the screen location and send the corresponding signal to the CPU.



Input Devices

- ❑ 3D interacting devices.
 - Cockpit and virtual controls
 - Steering wheels, knobs and dials...just like real!
 - Data glove
 - Fiber optics used to detect finger position
 - VR helmets
 - Detect head motion and possibly eyegaze



Output Devices

- ❑ Following are few of the input and output devices which are used in a computer.
 - Monitors
 - Speakers
 - Printers

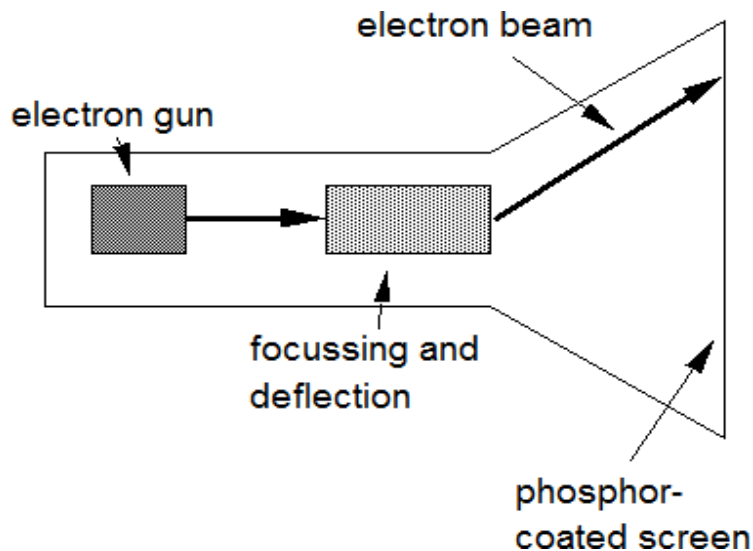
Output Devices

❑ Monitors

- Monitors, commonly called as Visual Display Unit (VDU), are the main output device of a computer.
- It forms images from tiny dots, called pixels that are arranged in a rectangular form. The sharpness of the image depends upon the number of pixels.
- There are two kinds of viewing screen used for monitors.
 - Cathode-Ray Tube (CRT)
 - Flat-Panel Display

Output Devices

- ❑ Cathode-Ray Tube (CRT) Monitors
- ❑ Stream of electrons emitted from electron gun, focused and directed by magnetic fields, hit phosphor-coated screen which glows.



Output Devices

❑ Flat-Panel Display Monitor

- The flat-panel display refers to a class video devices that have reduced volume, weight and power requirement in comparison to the CRT.



Output Devices

❑ Flat-Panel Display Monitor

This can be divided into two categories.

➤ Emissive Display

- The emissive display are devices convert electrical energy into light. E.g. Plasma panel and LED (Light Emitting Diodes)

➤ Non-Emissive Display

- The Non- emissive displays use optical effects to convert sunlight or light from some other source into graphic patterns. E.g. LCD (Liquid Crystal Device)

Output Devices

❑ Printers

- Printer is an output device, which is used to print information on paper.
- There are two types of printers
 - Impact Printers.
 - Non-Impact Printers.

Output Devices

❑ Impact Printers

- The impact printers print the characters by striking them on the ribbon which is then pressed on the paper.
- These are two types.
 - Character Printers.
 - Line Printers.

Output Devices

❑ Impact Printers

➤ Character Printers

- Print one character at a time.
- These are further divided into two types
 - Dot Matrix Printer
 - Daisy Wheel



Output Devices

- ❑ Impact Printers

- Line Printers

- Print one line at a time.
 - These are further divided into two types
 - Drum Printer
 - Chain Wheel



Output Devices

❑ Non-Impact Printers

- Print the character without using ribbon. These printers print a complete page at a time so they are also called as Page Printers.
- These are two types.
 - Laser Printers.
 - Inkjet Printers.

Output Devices

- ❑ Non-Impact Printers – Laser printers
 - User laser light to produce the dots needed to form the characters to be printed on a page.



Output Devices

- ❑ Non-Impact Printers – Inkjet printers
 - Print characters by spraying small drops of ink onto paper.



Output Devices

❑ Speakers

- Produce computer output as sound.
- They are common devices in computer system and examples include
 - Built-in speaker
 - Two-device set speakers
 - Surround sound speakers

THANK YOU